MACHINE TOOLS

Lathe • Milling • Shaper • Planer • Drilling • Grinding

Reference: Elements of Workshop Technology, Vol. II - Hajra Choudhury

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1. General Aspects of Machine Tools

A machine tool is a power-driven machine used to shape or finish metal or other engineering materials by controlled relative motion between the workpiece and a cutting or abrasive tool. Machine tools provide the required motions, power, rigidity, accuracy and means of holding the tool and work.

Term	Meaning
Primary cutting motion	Main motion that produces cutting, such as work rotation in a lathe or cutter rotation in milling.
Feed motion	Secondary controlled motion that brings new material into the cutting zone.
Tool holding	Arrangement used to hold and accurately position the cutting tool.
Work holding	Arrangement used to grip, support and locate the workpiece.
Attachment	Auxiliary device fitted to extend the range or capability of a basic machine tool.

2. Lathe Machine

The lathe is one of the most versatile machine tools. Its basic principle is to rotate the workpiece about its axis while a cutting tool is fed against it. The machine is mainly used for producing cylindrical, conical, threaded and faced surfaces.

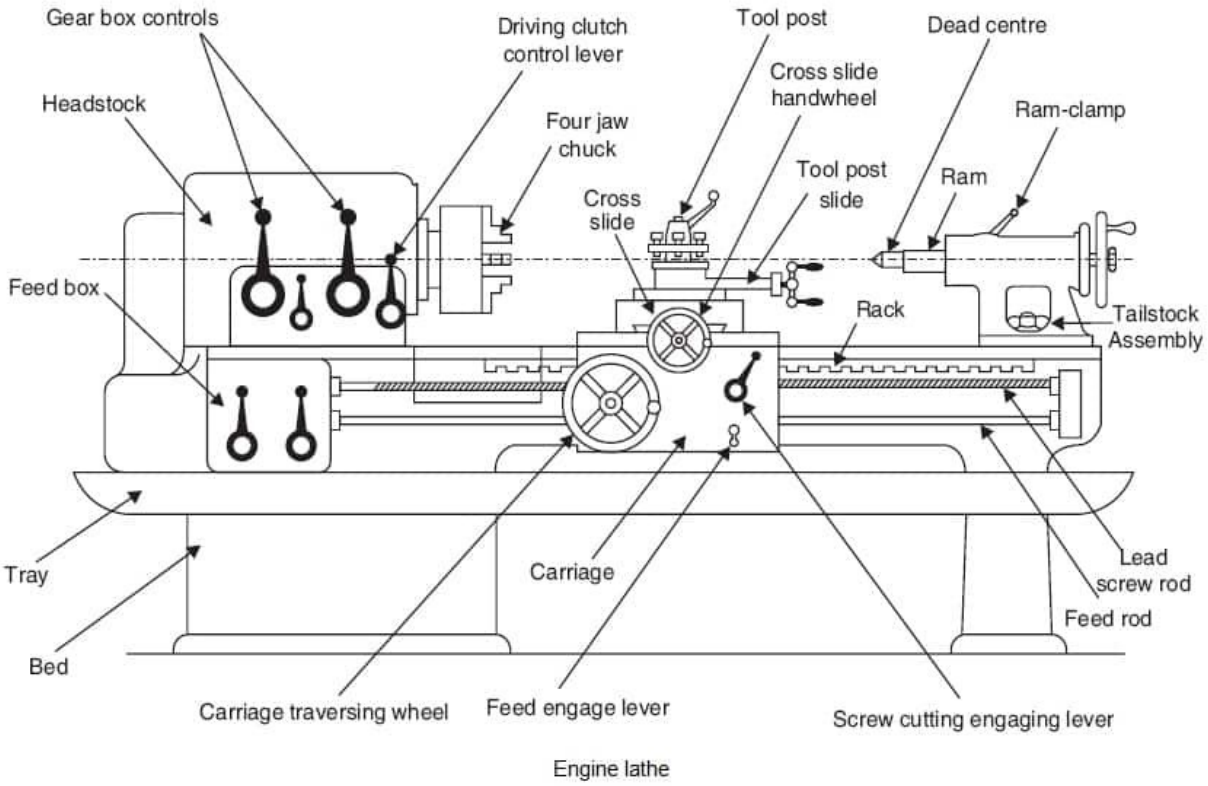


Figure. Main parts of an engine lathe.

Working principle

The work rotates in a spindle while the cutting tool is given longitudinal, cross or angular feed. Material is removed in the form of chips as the cutting edge engages the rotating work.

Functions / operations

- Straight turning
- Facing
- Taper turning
- Thread cutting
- Knurling
- Parting-off and grooving
- Drilling, boring and reaming using the tailstock
- Form turning and polishing

Types

- Speed lathe
- Engine or centre lathe
- Bench lathe
- Tool-room lathe
- Capstan and turret lathe
- Automatic lathe
- Special-purpose lathe
- CNC lathe / turning centre

Size and specifications

- Swing over bed

- Distance between centres
- Height of centres above bed
- Length and width of bed
- Spindle bore
- Range and number of spindle speeds
- Range of feeds
- Lead-screw pitch
- Motor power

Main parts

Part	Function
Bed	Rigid base carrying headstock, tailstock and carriage; contains guideways.
Headstock	Houses spindle, speed-change mechanism and driving system.
Spindle	Carries chuck, faceplate or other work-holding device and rotates the work.
Tailstock	Supports long work and holds drills, reamers and centres.
Carriage	Supports and moves cutting tool; includes saddle, cross-slide, compound rest and apron.
Lead screw / feed rod	Transmit power for thread cutting and automatic feeds.

Tool-holding devices

- Single-way tool post
- Four-way tool post
- Quick-change tool post
- Boring bar holder
- Tailstock drill chuck / taper socket

Work-holding devices

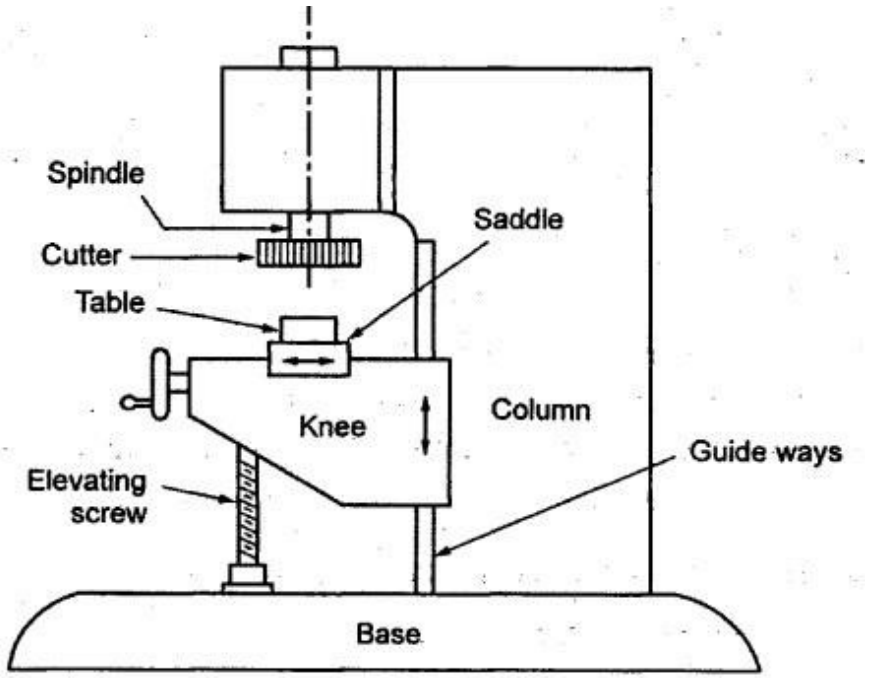
- Three-jaw self-centring chuck
- Four-jaw independent chuck
- Collet chuck
- Faceplate
- Lathe centres with driving plate and carrier
- Mandrels
- Steady rest and follower rest

Attachments / accessories

- Taper-turning attachment
- Milling attachment
- Grinding attachment
- Copying/tracer attachment
- Relieving attachment
- Thread chasing attachment

3. Milling Machine

Milling removes metal by means of a rotating multi-point cutter. The work is mounted on a table and fed against the rotating cutter. Milling is widely used for flat surfaces, slots, gears, profiles and contoured surfaces.



Working principle

The cutter rotates continuously while the work is fed past it. Each cutter tooth removes a small chip during its engagement with the work.

Functions / operations

- Plain/face milling
- Slot and keyway milling
- Angular milling
- Form milling
- Gear cutting
- Helical milling
- Profile and contour milling
- Straddle and gang milling

Types

- Column-and-knee milling machine: horizontal, vertical and universal
- Fixed-bed / manufacturing type
- Planer-type milling machine
- Rotary-table milling machine
- Special-purpose milling machines
- CNC machining centres

Size and specifications

- Table length and width
- Longitudinal, cross and vertical table travels
- Number/range of spindle speeds
- Feed range
- Spindle taper
- Distance from spindle to table
- Motor power
- Maximum work weight

Main parts

Part	Function
Base	Supports machine and often acts as coolant reservoir.
Column	Main vertical body housing drive and spindle mechanism.
Knee	Vertically adjustable casting supporting saddle and table.
Saddle	Provides cross movement.
Table	Holds work and provides longitudinal movement.
Spindle / arbor	Rotates milling cutter.
Overarm and arbor support	Support horizontal milling arbor.

Tool-holding devices

- Arbor for plain/side milling cutters
- Stub arbor
- Shell-end-mill arbor
- Collet chuck
- End-mill holder
- Drill chuck
- Morse/ISO taper adapters

Work-holding devices

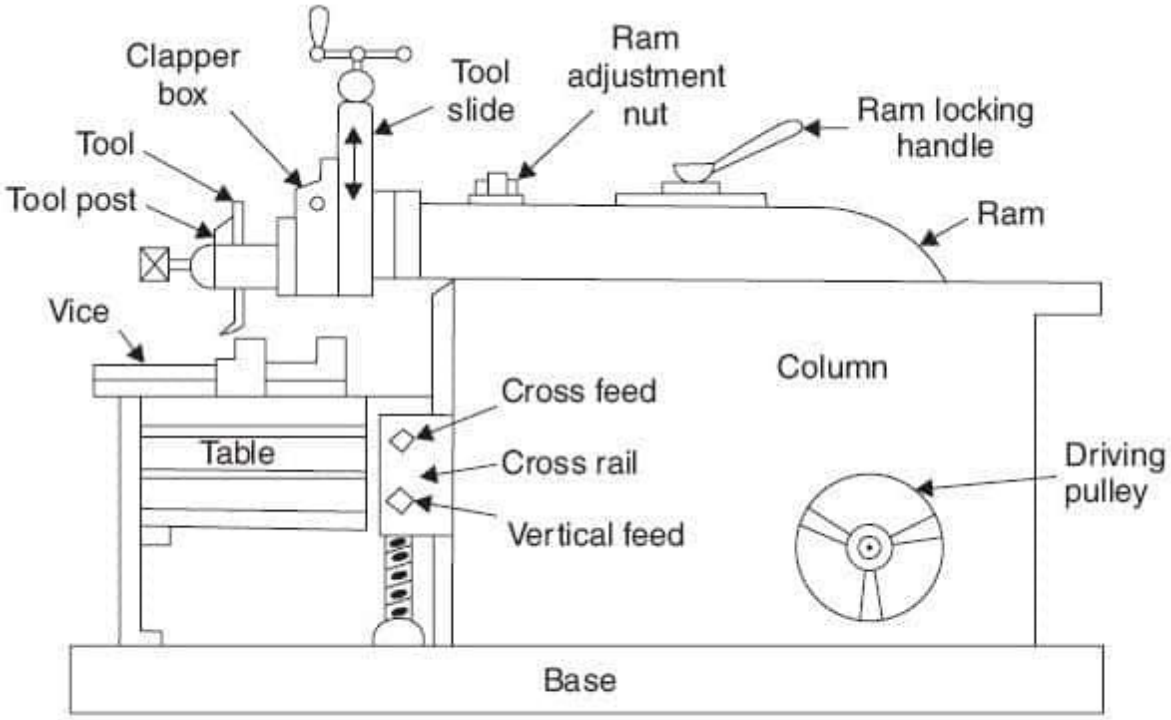
- Machine vice
- T-bolts, step blocks and clamps
- Angle plates
- V-blocks
- Dividing/indexing head
- Rotary table
- Special fixtures

Attachments / accessories

- Vertical milling attachment
- Slotting attachment
- Universal dividing head
- Rotary table
- High-speed attachment
- Rack-milling attachment
- Helical milling / universal table arrangement

4. Shaper Machine

A shaper produces flat, angular or contoured surfaces by reciprocating a single-point cutting tool over a stationary workpiece. Cutting normally occurs during the forward stroke and the return stroke is idle.



Principal parts of a shaper

Working principle

The ram carrying the tool reciprocates in a straight line. Feed is given to the work table, usually at the end of each return stroke. A quick-return mechanism reduces non-cutting return time.

Functions / operations

- Machining horizontal flat surfaces
- Vertical surfaces
- Inclined surfaces
- Slots and grooves
- Keyways
- Dovetails
- Irregular contours using form tools or special setups

Types

- Crank shaper
- Geared shaper
- Hydraulic shaper
- Horizontal shaper
- Vertical shaper / slotter
- Travelling-head shaper

Size and specifications

- Maximum length of ram stroke
- Maximum table travel
- Table size
- Maximum distance from ram to table
- Number of ram strokes per minute
- Feed range
- Motor power

Main parts

Part	Function
Base	Heavy foundation of the machine.
Column	Houses driving mechanism and guides the ram.
Ram	Reciprocating member carrying tool head.
Tool head / clapper box	Holds tool and permits lifting during return stroke.
Cross-rail	Supports saddle and table; adjustable vertically.
Table	Carries vice or work; receives cross feed.

Tool-holding devices

- Tool post on clapper box
- Straight and bent tool holders
- Boring tool holder
- Form-tool holder

Work-holding devices

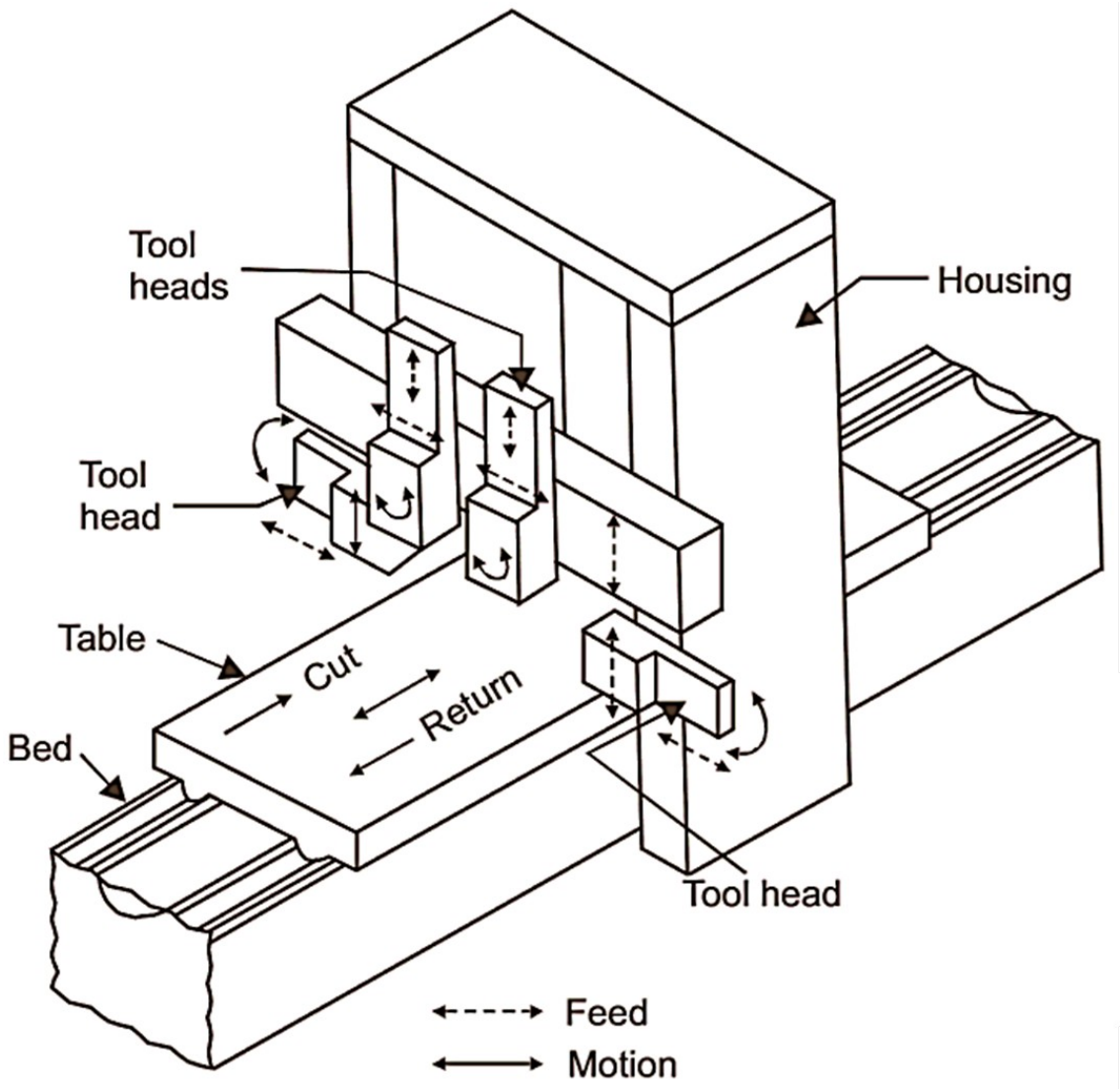
- Shaper vice
- T-bolts and clamps
- Angle plates
- V-blocks
- Special fixtures

Attachments / accessories

- Universal vice
- Rotary table/indexing attachment
- Special contour/form attachments
- Slotting-type attachments on special machines

5. Planer Machine

A planer is intended mainly for machining large and heavy workpieces. Unlike a shaper, the work reciprocates on a table while one or more tools remain in tool heads mounted on cross-rails or columns.



(a) Block diagram showing the basic components of a double-housing planer

Working principle

The workpiece is fixed on the reciprocating table. Cutting takes place during one direction of table travel. Feed is given to the tool head between strokes.

Functions / operations

- Machining large flat surfaces
- Machining multiple parallel surfaces in one setting
- Guideways and machine beds
- Vertical and angular surfaces
- Long slots and grooves
- Large structural and fabrication components

Types

- Double-housing planer
- Open-side planer
- Pit planer

- Edge/plate planer
- Divided-table planer
- Special-purpose planer

Size and specifications

- Maximum table length and width
- Maximum work height under cross-rail
- Table travel/stroke length
- Number of tool heads
- Table speed range
- Cross and vertical feed range
- Maximum work weight
- Motor power

Main parts

Part	Function
Bed	Long rigid foundation with accurately machined guideways.
Table	Supports and reciprocates with the workpiece.
Columns / housings	Support cross-rail and side tool heads.
Cross-rail	Carries upper tool heads; adjustable vertically.
Tool heads	Hold tools and provide feed.
Drive and reversing mechanism	Provides controlled table reciprocation.

Tool-holding devices

- Tool posts in rail heads
- Side tool-head holders
- Heavy-duty single-point tool holders
- Boring/slotting holders where provided

Work-holding devices

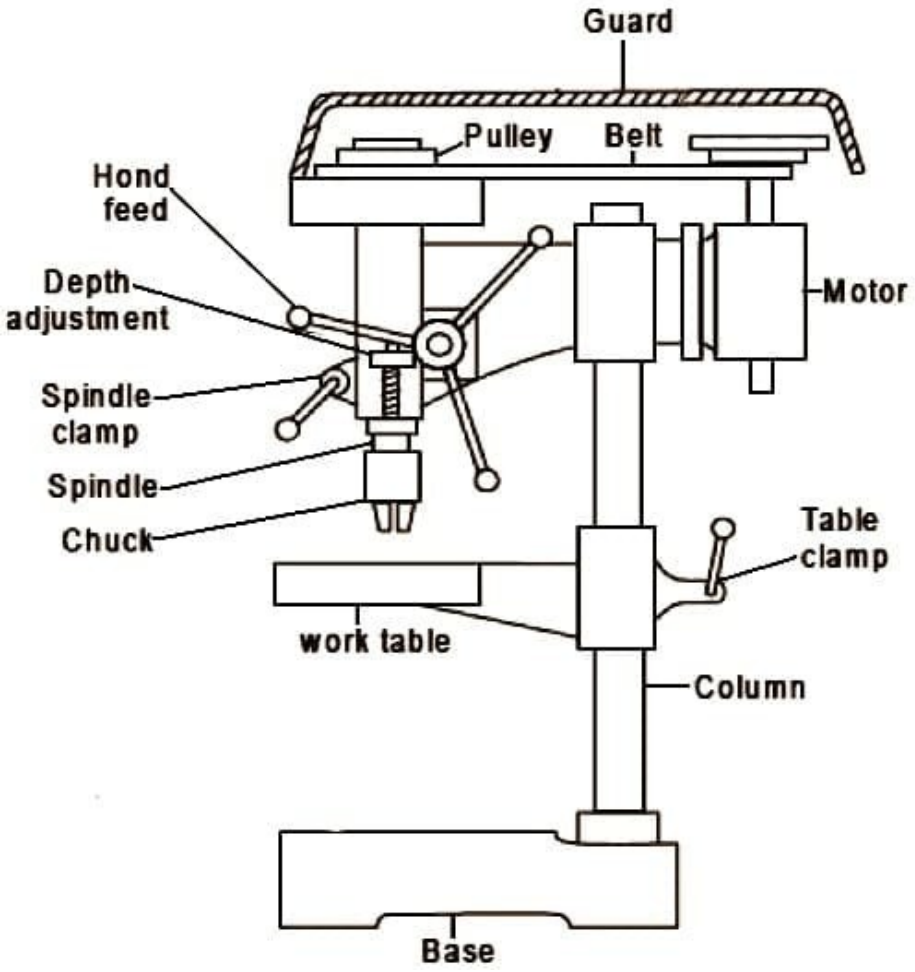
- T-bolts and heavy clamps
- Stops and packing blocks
- Angle plates
- V-blocks
- Special large fixtures

Attachments / accessories

- Multiple tool heads
- Side tool heads
- Special milling/grinding heads on combination machines
- Copying or contouring devices on special planers

6. Drilling Machine

A drilling machine primarily produces circular holes using a rotating drill. The tool rotates and is fed axially into a stationary workpiece. The machine can also perform several related hole-finishing operations.



Working principle

The drill rotates about its axis while axial feed pushes it into the work. The two cutting lips remove material and chips move out through the drill flutes.

Functions / operations

- Drilling
- Reaming
- Boring
- Counterboring
- Countersinking
- Spot facing
- Tapping
- Trepanning

Types

- Portable drilling machine
- Sensitive/bench drilling machine
- Upright/pillar drilling machine
- Radial drilling machine
- Gang drilling machine
- Multi-spindle drilling machine
- Deep-hole drilling machine
- CNC drilling centre

Size and specifications

- Maximum drilling capacity
- Spindle-to-column distance
- Maximum spindle travel
- Spindle speed range
- Feed range
- Table size and movement
- Spindle taper
- Motor power

Main parts

Part	Function
Base	Supports machine and work.
Column	Vertical support for head/table.
Drill head	Contains spindle drive and feed mechanism.
Spindle / quill	Rotates and feeds cutting tool.
Table	Supports work-holding devices.
Feed mechanism	Provides hand or power feed.

Tool-holding devices

- Drill chuck
- Morse taper socket
- Sleeves and sockets
- Collet chuck
- Tap holder
- Arbor

Work-holding devices

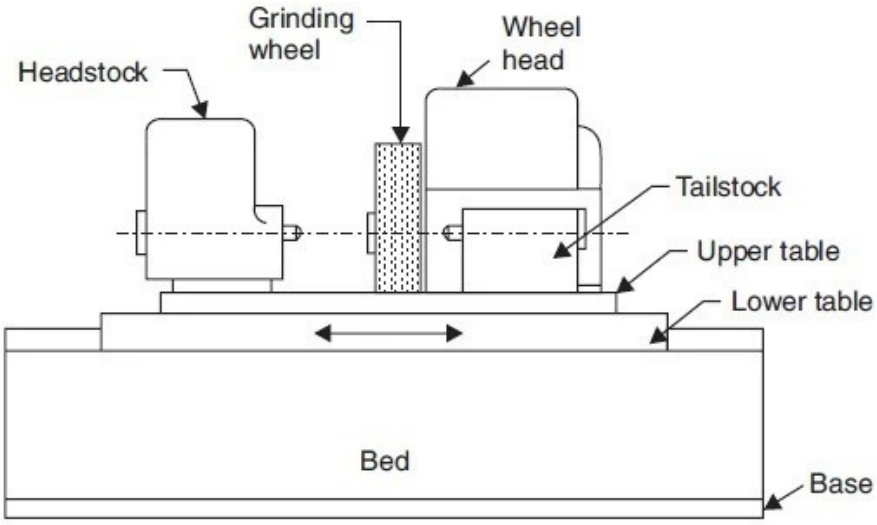
- Drill vice
- T-bolts and clamps
- V-blocks
- Angle plates
- Jigs and fixtures

Attachments / accessories

- Tapping attachment
- Multi-spindle head
- Right-angle drilling attachment
- Sensitive feed attachment
- Depth stop
- Coolant attachment

7. Grinding Machine

Grinding is an abrasive machining process in which a rotating grinding wheel removes very small chips by the action of numerous hard abrasive grains. It is used for high dimensional accuracy, close tolerances and fine surface finish.



Block diagram of a plain cylindrical grinder

Working principle

The grinding wheel rotates at high peripheral speed. Abrasive grains act as tiny cutting edges. Relative feed between wheel and work removes a thin layer of material.

Functions / operations

- Surface grinding
- Cylindrical external grinding
- Internal grinding
- Centreless grinding
- Tool and cutter sharpening
- Form/profile grinding
- Finishing hardened components

Types

- Bench/pedestal grinder
- Surface grinding machine
- Cylindrical grinding machine
- Internal grinder
- Centreless grinder
- Tool and cutter grinder
- Special-purpose and CNC grinders

Size and specifications

- Maximum work/table size
- Grinding wheel diameter, width and bore
- Wheel spindle speed
- Table travel
- Maximum centre height and distance between centres for cylindrical grinders
- Feed range
- Motor power

Main parts

Part	Function

Bed/base	Provides rigid support and guideways.
Wheel head	Carries grinding-wheel spindle and drive.
Grinding wheel	Abrasive cutting tool.
Work table / carriage	Provides work movement in surface grinding.
Headstock and tailstock	Support rotating work in cylindrical grinding.
Coolant system	Supplies fluid to control heat and remove swarf.

Tool-holding devices

- Grinding-wheel spindle and flanges
- Wheel adapters
- Wheel guards
- Tool rests on bench grinders
- Special arbors for internal grinding wheels

Work-holding devices

- Magnetic chuck for ferromagnetic flat work
- Mechanical chuck
- Centres and driving dog
- Collets
- Faceplate
- V-blocks and fixtures
- Work-rest blade in centreless grinding

Attachments / accessories

- Internal grinding attachment
- Radius/form dressing attachment
- Wheel balancing attachment
- Coolant and filtration system
- Digital gauging / in-process sizing devices

8. Shaper versus Planer

Point	Shaper	Planer
Reciprocating member	Tool/ram reciprocates	Work/table reciprocates
Work size	Small to medium	Large and heavy
Number of tools	Usually one tool	Several tool heads can cut simultaneously
Cutting speed	Lower power and lighter cuts	Higher power and heavier cuts
Feed	Generally given to work/table	Generally given to tool heads
Construction	Comparatively light and compact	Heavy, rigid and large
Typical use	Small flat surfaces, slots, keyways	Machine beds, large plates and long guideways

9. Important Points

- Lathe: work rotates and the single-point tool feeds against it.
- Milling: multi-point cutter rotates while the work is fed relative to it.
- Shaper: tool reciprocates; work receives feed.
- Planer: work reciprocates; one or more tools receive feed.
- Drilling: drill rotates and feeds axially into the stationary work.
- Grinding: a high-speed abrasive wheel removes very small chips and produces fine finish.
- Machine-tool specifications normally include work capacity, travel, speed/feed range, spindle details and motor power.
- Correct work holding is essential for rigidity, accuracy and safety.
- Attachments extend the capability of the basic machine without changing its fundamental principle.

Reference

Hajra Choudhury, S. K. and Hajra Choudhury, A. K., Elements of Workshop Technology, Vol. II.